

Concept Learning

Introduction to Boolean-Valued Concept Learning

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This is exactly what our learning algorithms aim to do.

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Concept learning =

Learning this YES/NO function from positive and negative examples.

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- WeaponMarks
- ItemsMissing
- NoiseReports
- WitnessReports

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“Is this location a crime scene?”

Each location is described by attributes such as:

- DamageType
- BloodTraces
- WeaponMarks
- ItemsMissing
- NoiseReports
- WitnessReports

Each example is labeled as:

- **CrimeScene = Yes** (positive)
- **CrimeScene = No** (negative)

Training Examples

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A small dataset may look like this:

Ex	Damage	Blood	Marks	Missing	Noise	Witness	Crime
1	ForcedEntry	No	No	Yes	None	No	Yes
2	BrokenGlass	Yes	Yes	No	Gunshot	Yes	Yes
3	None	Yes	No	No	None	No	No
4	ForcedEntry	Yes	Yes	Yes	None	Yes	Yes

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4	ForcedEntry	Yes	Yes	Yes	None	Yes	Yes

Goal: learn a rule that predicts the label for any **new** location.

Hypothesis Representation

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An example hypothesis:

(ForcedEntry, ?, ?, Yes, ?, ?)

represents: *“A location is a crime scene only if there is forced entry and items missing.”*

Meaning of Symbols: $?$, Specific Value, $\emptyset$

Specific Value

- Attribute must exactly match
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Example:

$(\emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset)$

classifies **no** instance as a crime scene.

Most General and Most Specific Hypotheses

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Most Specific Hypothesis

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Rejects every location.

Learning algorithms search between these two extremes to find the best hypothesis.

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Goal: Find $h \in H$ such that $h(x) = c(x)$

for all instances.

2.2.1 Notation: The Building Blocks of Concept Learning

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To formalize concept learning, we define a few essential components.

Instance Space (X):

- The set of all possible examples the learner may encounter
- In our crime-scene scenario, each location is described using attributes:
 - DamageType (ForcedEntry, BrokenGlass, None)
 - BloodTraces (Yes, No)
 - WeaponMarks (Yes, No)
 - ItemsMissing (Yes, No)
 - NoiseReports (Gunshot, None)
 - WitnessReports (Yes, No)

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- The true function the learner must discover
- A boolean-valued function:

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- $c(x) = 1$: location *is* a crime scene
- $c(x) = 0$: location *is not* a crime scene
- The learner **does not know** this function and must infer it

Training Examples D

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- A subset of instances for which the correct label is known
- Positive examples: confirmed crime scenes
- Negative examples: verified non-crime cases
- These are the only facts available about the true concept

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All locations are crime scenes.

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Meaning:

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Most General Hypothesis:

(?, ?, ?, ?, ?, ?)

All locations are crime scenes.

Most Specific Hypothesis:

($\emptyset$, $\emptyset$, $\emptyset$, $\emptyset$, $\emptyset$, $\emptyset$)

No location is a crime scene.

Goal of Concept Learning

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such that

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Because we cannot observe all of X , we rely on a powerful assumption about learning. . .

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This is the fundamental foundation of all learning methods.

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- Teachers could not predict student performance
- Humans could not learn language
- Machine learning would be impossible

Example: Why Induction Works

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Case	ForcedEntry	ItemsMissing	Blood	Crime?
1	Yes	Yes	No	Yes
2	Yes	Yes	Yes	Yes
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Learner forms rule:

$\text{ForcedEntry} = \text{Yes} \Rightarrow \text{CrimeScene} = \text{Yes}$

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Learner forms rule:

$$\text{ForcedEntry} = \text{Yes} \Rightarrow \text{CrimeScene} = \text{Yes}$$

Because all observed positive cases share this pattern.

Summary

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- c : true concept
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Induction explains:

- Why learning from examples is possible
- Why generalization to unseen data works

EnjoySport Concept Learning Task (Table 2.2 Format)

Given:

- **Instances X:** Possible days described by attributes:
 - Sky: Sunny, Cloudy, Rainy
 - AirTemp: Warm, Cold
 - Humidity: Normal, High
 - Wind: Strong, Weak
 - Water: Warm, Cool
 - Forecast: Same, Change
- **Hypotheses H:** Conjunction of constraints over these attributes. Each constraint may be:
 - “?” $\rightarrow$ any value acceptable
 - “ $\emptyset$ ” $\rightarrow$ no value acceptable
 - specific value (e.g., Warm)
- **Target concept:**

$$c : \text{EnjoySport} : X \rightarrow \{0, 1\}$$

- **Training examples D:** Positive and negative labeled examples from Table 2.1

Lecture Summary

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Key Insight: Inductive Learning Hypothesis

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Why this matters:

- Without induction, no human or machine could generalize from